

How Can a Machine Learning Classification Model Assist in Determining the Magnitude of a Rotating Shaft's Unbalance?

Ashar Siddiqui
Santa Clara High School
Santa Clara, CA, USA

Abstract—This paper investigates the extent to which a Random Forest (RF) classification machine learning model can assist in determining the magnitude of unbalance in a rotating shaft. Using vibration analysis data recorded experimentally by Mey et al. at the Fraunhofer Institute for Integrated Circuits, an RF classification model was trained and tested on datasets corresponding to the two most extreme unbalance levels: level “0” (no unbalance) and level “4” (maximum unbalance). The model achieved 100% accuracy on both training and testing subsets, a result that is well supported by several factors: the high distinguishability of the two extreme unbalance cases, the large size of the dataset, and the effectiveness of the statistical features, such as standard deviation and kurtosis, extracted from the vibration sensor data during preprocessing. These findings suggest strong potential for implementing RF classification models in industrial predictive maintenance systems, though future work should extend testing to intermediate unbalance levels “1”, “2”, and “3” to more fully validate the model’s generalizability.

Index Terms—rotating shaft, unbalance detection, Random Forest, machine learning, predictive maintenance, vibration analysis

I. INTRODUCTION

In the dynamic and ever changing landscape of the modern industrial world, the role of machinery and other mechanical parts is ever persistent and crucial. Mechanical parts and their individualistic functionality allow for the effective production in a variety of fields and areas, ranging from aerospace manufacturing to medical biotechnology. The maintenance and prolongation of a mechanical part is also a crucial part of industrial progress, as the properly functioning aspect of machinery is a determinant of the economic viability, industrial success, and working safety of any industrial business that employs such machinery to contribute to their production and output. It was reported that 99% of mechanical failures are preceded by noticeable indicators [1]. Awina Heng, a researcher affiliated with Queensland University of Technology, believes that in recent years, technologies in machine condition monitoring (CM) and fault diagnostics have become more developed, and data such as vibration signatures and acoustic emissions signatures processed and analyzed through state-of-the-art sensors and computation technologies. Thus, determining such noticeable indicators and being able to determine when such parts need to be replaced or serviced early on can vastly contribute to industry output and efficiency.

As a result, it could be beneficial for factory owners to detect defects early, as to minimize cost and time repercussions. However, it is also important to approach the maintenance of mechanical parts in a cautionary and strategic method, in order to effectively diagnose and service such parts accordingly.

II. CONTEXT

The 21st century has witnessed an immense surge in the popularity, adoption, and implementation of mechanical parts and machinery. This wide adoption is attributed to a variety of factors, including optimistic advancements in technology, sustainability efficiency, and scaling and miniaturization of machinery. Regarding advancements in technology, the 21st century alone has seen an impressive rapid development in automation and artificial intelligence. As various forms of artificial intelligence are being implemented in existing technology such as vehicular transportation, the demand for mechanical parts has skyrocketed. A report published by The American Society of Mechanical Engineers (ASME) entitled “Mechanical Engineering in the 21st Century: Trends Impacting the Profession” prepared by the Hudson Institute, Inc. (1999) explains the evolution of mechatronics from the 20th to 21st century [2]. Specifically, it categorizes the trends of change in engineering as technological change, demographic change, economic change and social change. Not only is the demand for mechanical parts rising exponentially, but the need for such parts to be effective and efficient is also crucial.

Perhaps one of the most significant and important mechanical parts needed to be effective and efficient is rotating machinery. Specifically, a rotating part or rotating piece of machinery is a cylindrical component that rotates around its axis. Being used in applications where power is transmitted from one part to another, rotating shafts are crucial in aiding this process. S.A.A. Hosseini of the Department of Mechanical Engineering at Tarbiat Modarres University explains that rotating shafts are used for power transmission in many modern machines [3]. Furthermore, while in use, rotating shafts tend to bend when at certain speeds, and whirl at a rapid pace. The whirling output of such a part is primarily caused by the rotating shaft’s mass unbalance and friction. As a result, when rotating shafts are used extensively, their unbalance tends to increase, indicating possible defects.

With the advent of artificial intelligence in the workspace, more attention has been focused on the extent to which it can assist humans in their tasks. One particular use of artificial intelligence in the work environment is its ability to contribute to the field of predictive maintenance. The application of Internet of Things, i.e. smart devices and sensors, increases collection and availability of diverse data. Predictive maintenance provides a detailed examination of the detection, location and diagnosis of faults in related machineries using various analyses [4]. Forms of artificial intelligence like machine learning models, predictive algorithms, and data analytics now have the potential to transform traditional predictive maintenance methods. As a result, artificial intelligence promises a more effective, efficient, and optimized approach to predictive maintenance.

III. CURRENT RESEARCH PERSPECTIVES

Predictive maintenance, in academia, has several definitions. One of its most fundamental definitions is the monitoring of the vibration of rotating machinery in an attempt to detect incipient problems and to prevent catastrophic failure [5]. However, the more common, agreed upon notion of predictive maintenance is that it describes the regular monitoring of the mechanical condition, operating efficiency, and effectiveness of a certain mechanical part of a particular machine. R. Keith Mobley, a self-employed executive advisor and expert in the field of Operating Dynamics Modeling and Predictive Analytics & Risk Management, explains that predictive maintenance is “the means of improving productivity, product quality, and overall effectiveness of manufacturing and production plants” [5]. Mobley does not believe that predictive maintenance is confined and restricted to terms such as “thermal imaging” or “lubricating oil analysis” that market the idea of predictive maintenance as a tool. Instead, he believes that it is a mindset that employs the most effective and efficient tools in order to directly monitor the mechanical condition and system performance of a machine-train and system in the plant in order to improve the product quality, output, and profitability of manufacturing in an industrial business environment.

The principles of predictive maintenance are also explained in depth by Cornelius Scheffer, who holds a PhD in Mechanical Engineering and is a practical professional in the area of Practical Machinery Vibration Analysis and Predictive Maintenance [6]. Scheffer explains that predictive maintenance monitors mechanical condition, equipment efficiency and other parameters and attempts to derive the approximate time of a functional failure. Some notable predictive maintenance techniques mentioned in Scheffer’s book titled “Practical Machinery Vibration Analysis and Predictive Maintenance” include vibration analysis, oil and wear debris analysis, ultrasonics, thermography, and performance evaluation. In addition to this, Scheffer utilizes a medical analogy in order to describe the notion of predictive maintenance simply. In essence, Scheffer explains that whenever a human body has an issue, it exhibits a symptom. The nervous system provides the information, which is the detection state. Furthermore, if required, pathological

tests are done to diagnose the problem. On this basis, suitable treatment is recommended. In a similar way, defects that occur in a machine always exhibit a symptom in the form of vibration or some other parameter [6]. It is at this point that predictive maintenance techniques are incorporated. Specifically, techniques like vibration analysis detect the symptoms of the mechanical part’s defects, and then assist in diagnosing the defects and estimating the severity of the particular defects.

The notions and sentiment in regards to the idea that machine learning and artificial intelligence technology should be implemented into the field of predictive maintenance is shared by many. Mounia Achouch, a computer science researcher from the Centre for Research and Innovation in Energy Intelligence from the University of Quebec at Rimouski, explains that predictive maintenance “is a recent preventive maintenance approach that consists of improving the performance and efficiency of the manufacturing process by increasing the lifespan of equipment and ensuring sustainable operational management” [7]. Achouch believes that successful predictive maintenance can result in a decrease in downtime and the number of unnecessary stops, as well as a reduction in repair costs. Big data, new devices, and controlled production have resulted in the fruition of Industry 4.0, which implies the need to apply automation techniques to integrate all the new technologies that will subsequently increase productivity [7]. This will allow such businesses to keep pace with competitors, as well as boost productivity and output.

However, the sentiment and idea that machine learning and artificial intelligence technology should be implemented into the field of predictive maintenance is not shared by all. Pedro Nunes, a researcher based in the Department of Mechanical Engineering, University of Aveiro, believes that anomaly detection, one of the most common implementations of AI-based predictive maintenance, has many challenges related to the untypical nature of data in the field [8]. This data comes from simulated environments where running mechanical parts are connected to sensors and produce numerical data that reveal their mechanical age and efficiency. While an anomaly caused by a machine malfunction or degradation contains valuable information, anomalies caused by a sensor’s malfunction, low battery, or other external disturbance are considered noise and may cause misinterpretations [8]. Due to the fact that inaccurate data may be present in such simulations, Nunes believes that such inconsistencies may result in imprecise measurements of the efficiency of a certain mechanical part, thus resulting in an incorrect assessment of the part.

Delving into the role of machine learning in predictive maintenance, Dr. Zeki Murat Cinar, a research engineer at the Department of Mechanical Engineering in Eastern Mediterranean University, has determined that the key distinctions between supervised and unsupervised learning would result in different success rates in their implementations in predictive maintenance [9]. Cinar explains that in unsupervised machine learning, there is no feedback from an external teacher or knowledgeable expert. Based on the existing data, the algorithm identifies the clusters. Supervised learning’s main goal

is determining the unknown classes of items by clustering, where classification is most common for supervised learning cases. Cinar also explains that unsupervised ML basically defines any ML method that attempts to learn structure in the absence of either an identified output (like supervised ML) or feedback (like Reinforcement Learning (RL)) [9]. Cinar identifies clustering, self-organizing maps, and association rules to be the three main examples of unsupervised learning. Overall, it is clear to see that there does exist heavy debate on the certain type of machine learning model to implement into predictive maintenance, as each model has different benefits and limitations. Furthermore, there is no solitary machine learning model that has been completely adopted by industry in regards to predictive maintenance, so identifying the extent to which a particular type of classification machine learning model is successful in predictive maintenance could prove to be very beneficial to industrial businesses.

IV. FINAL JUSTIFICATION

Regarding the machine learning model that would be used to determine and predict such defects, a Random Forest classification algorithm was used. Specifically, classification is a supervised machine learning concept in which the algorithm categorizes input. According to Adam Shafi, a Data Scientist at Deliveroo, in a random forest classification, multiple decision trees are created using different random subsets of the data and features. Each decision tree is like an expert, providing its opinion on how to classify the data. Predictions are made by calculating the prediction for each decision tree, then taking the most popular result [10]. A Random Forest classification algorithm was chosen due to the fact that it incorporates supervised learning. Specifically, due to the fact that predictive maintenance data exists primarily in the form of classification data, a Random Forest classifier would work most effectively in this case. During this research, a Random Forest classification algorithm was used to classify the stage of unbalance of a particular rotating shaft. The effects of unbalance of rotating shafts can result in decreased life of bearings, unsafe work conditions, and overall reduced machine life. Determining whether such a mechanical part in industry is defective or not can prove to be particularly beneficial, especially when such defects can be uncovered early on.

The use and methods of machine learning models can be implemented in this area to determine the magnitude of a rotating shaft's unbalance. Trained on a dataset in which rotating shafts have been exposed to vibration, yielding a certain unbalance level, this machine learning model can evaluate the unbalance level of a rotating shaft given a particular rotating shaft's input vibration data. Finally, depending on the unbalance level, the machine learning model will be able to predict whether a particular rotating shaft requires maintenance. The deterioration of rotating shafts has resulted in such shafts completely failing in their implementation. Several potential causes for the failure of shafts include corrosion, rotation at excessive speeds, overload, and fatigue. For the purpose of this research paper, the cause that will be focused on is the

unbalance of a rotating shaft. Rotating unbalance involves the uneven rotation of mass along a certain axis of the shaft. The unbalance level and unevenness are directly proportional, and the unbalance level can be measured from a scale of 1–4, where 1 is the least unbalanced and 4 is the most unbalanced.

The unbalance level of a rotating shaft is determined when vibration signals are created from the rotating shaft. A machine learning classification algorithm would be appropriate when approaching this problem, as a classification approach would attempt to classify the unbalance level of a particular rotating shaft given vibration signals that were used to measure the level of unbalance on test rotating shafts. As a result, such a classification model can be given new vibration signal data and predict the unbalance level of the new rotating shaft. In this way, the algorithm is able to “learn” from existing data, and then produce a result that would help it determine whether a particular rotating shaft requires maintenance (if unbalance level is 3 or 4). Thus, the unbalance of a rotating shaft directly correlates with the rotating shaft's defectiveness, and classifying the unbalance level of a particular rotating shaft would prove to be greatly beneficial, as it would allow for such defects to be classified and determined at an early stage, resulting in the minimizing of time and cost repercussions.

V. RESEARCH QUESTION

As rotating shafts are extremely common in industry and are commonly subject to degradation and malfunction due to their excessive use, these particular types of mechanical parts have been chosen as the focus of this project. Furthermore, to assist predictive maintenance, a Random Forest Classification machine learning model has been chosen to be implemented, due to its supervised learning nature and ability to classify unbalance levels of a rotating shaft based on existing inputted data. Additionally, since there is no solitary machine learning model that has been completely adopted by industry in regards to predictive maintenance, identifying the extent to which a particular type of classification machine learning model is successful in predictive maintenance could prove to be very beneficial to industrial businesses. Ultimately, this paper focuses on the extent to which a Random Forest classification machine learning model is successful and suitable for predictive maintenance on a rotating shaft.

VI. METHODOLOGY

In order to determine how a machine learning classification model could assist in determining the magnitude of a rotating shaft's unbalance, it was necessary and essential to first identify an appropriate dataset to use. As mentioned previously, the use of vibration analysis is one of the most fundamentally sound and effective methods of predictive maintenance. Defects that occur in a machine always exhibit a symptom in the form of vibration or some other parameter [6]. Techniques like vibration analysis detect the symptoms of the mechanical part's defects, and then assist in diagnosing the defects and estimating the severity of the particular defects. Furthermore,

according to Mobley, one of predictive maintenance's fundamental aspects is the monitoring of the vibration of rotating machinery in an attempt to detect incipient problems and to prevent catastrophic failure [5]. Thus, since vibration analysis is a widely agreed upon technique to initially record and determine faults within rotating mechanical systems, seeking vibration analysis data was the primary step.

After conducting a search on Kaggle.com, a popular repository containing thousands of datasets to assist machine learning projects, a suitable dataset was identified under a Creative Commons Attribution 4.0 license [11]. Specifically, the dataset was created experimentally by Oliver Mey, Willi Neudeck, Andre Schneider, and Olaf Enge-Rosenblatt, a group of researchers at the Fraunhofer Institute for Integrated Circuits Division Engineering of Adaptive Systems. In their research paper titled "Machine Learning-Based Unbalance Detection of a Rotating Shaft Using Vibration Data," they explain their experimental process to gather vibration data from a rotating shaft [12].

The setup for the simulation of defined unbalances and the measurement of the resulting vibrations was powered by an electronically commutated DC motor (WEG GmbH, type UE 511 T), which was controlled by a motor controller (WEG GmbH, type W2300) and was fixed to the aluminum base plate by means of a galvanized steel bracket. Vibration sensors (PCB Synotech GmbH, type PCB-M607A11/M001AC) were attached to both the bearing block and the motor mounting and were read out using a 4-channel data acquisition system (PCB Synotech GmbH, type FRE-DT9837).

Using this setup, vibration data for unbalances of different sizes was recorded at a sampling rate of 4096 values per second. By varying the level of unbalance, different levels of difficulty were achieved, since smaller unbalances influenced the signals at the vibration sensors to a lesser extent. In total, datasets for 4 different unbalance strengths were recorded as well as one dataset with the unbalance holder without additional weight (i.e., without unbalance). The rotation speed was varied between approximately 630 and 2330 RPM in the development datasets and between approximately 1060 and 1900 RPM in the evaluation datasets. Each dataset is provided as a CSV file with five columns: (1) *V_{in}*: the input voltage to the motor controller (in V); (2) *Measured_RPM*: the rotation speed of the motor (in RPM); (3) *Vibration_1*: the signal from the first vibration sensor; (4) *Vibration_2*: the signal from the second vibration sensor; and (5) *Vibration_3*: the signal from the third vibration sensor.

Separate measurements were taken for each unbalance strength to enable a comparable division into development and evaluation datasets. The CSV files were named in the form XY.csv, where the digit X describes the unbalance strength ("0" = no unbalance, "4" = strong unbalance) and the letter Y describes the intended use ("D" = development/training, "E" = evaluation/testing). For example, 2D.csv refers to training data of unbalance level 2, and 4E.csv refers to testing data of unbalance level 4.

This dataset was chosen largely due to its categorical nature.

Since vibration data was categorized into 5 unbalance levels ("0" through "4"), this aligned well with the intended use of a Random Forest Classification model, which works most effectively with categorical data.

Upon finding the vibration analysis dataset, data was downloaded in the form 0D.csv, 0E.csv, 1D.csv, 1E.csv, 2D.csv, 2E.csv, 3D.csv, 3E.csv, 4D.csv, 4E.csv. Python 3 was used as the programming environment, with the tsfresh package installed to compute a wide range of features describing time series [12]. Since the first two columns (input voltage and rotation speed) were not of interest to this research, a preprocessing script was written to extract and batch process the relevant vibration sensor columns from each CSV file. The script reads each file, skips the first 50,000 rows corresponding to motor startup, and groups the remaining data into batches of 4096 rows. For each batch, statistical features including the mean RPM, and the standard deviation and kurtosis of each of the three vibration sensors, are computed to produce a compact "representative summary" of that batch. The full preprocessing script is shown below.

```
from os import extsep, mkdir
from os.path import dirname, join, exists
import sys
import numpy as np
import scipy.stats as st

def main():
    current_directory = dirname(__name__)
    kaggle_data_filenames = [
        fi for i in range(5)
        for letter in ('D', 'E')
        if exists(join(current_directory,
            (fi:=f'{i}{letter}')) + f'{extsep}csv')
    ]
    print(f'Data files found from Kaggle:
{kaggle_data_filenames}')
    directory_for_processed_data = join(
        current_directory, f'data')
    if (len(kaggle_data_filenames) != 0
        and not
        exists(directory_for_processed_data)):
        print(f'Creating directory ')

        f'"{directory_for_processed_data}"...'
        mkdir(directory_for_processed_data)
    for filename in kaggle_data_filenames:
        input_filename = join(
            current_directory,
            f'{filename}{extsep}csv')
        output_filename = join(
            directory_for_processed_data,
            f'{filename}{extsep}np')
        if not exists(output_filename):
            np.save(output_filename,
                process(input_filename),
                allow_pickle=False)
        else:
            print(
                f'The file "{output_filename}"
                already exists!'
                f' Please delete to override.',
                file=sys.stderr,
            )

def process(filename, batchsize=4096,
    skiprows=50000):
    """
```

```

Reads a CSV file as a 2D matrix. The header and
first
'skiprows' rows (motor startup) are ignored.
Rows are
grouped into batches of 'batchsize', and
statistics are
computed across each batch to produce a
representative
summary vector.
"""
print(f'Loading {filename}...')
d = np.genfromtxt(filename, delimiter=',',
    skip_header=1 + skiprows, dtype=np.float32)
numrows = d.shape[0]
truncated_numrows = numrows // batchsize *
    batchsize
d = d[:truncated_numrows, :]
d = d.reshape(truncated_numrows // batchsize,
    batchsize, -1)
rpm = d[:, 1, :]
vibration1 = d[:, 2, :]
vibration2 = d[:, 3, :]
vibration3 = d[:, 4, :]
print(f'Processing {d.shape[0] * d.shape[1]}
    rows...')
return np.stack([
    np.mean(rpm, axis=-1),
    np.std(vibration1, axis=-1),
    st.kurtosis(vibration1, axis=-1),
    np.std(vibration2, axis=-1),
    st.kurtosis(vibration2, axis=-1),
    np.std(vibration3, axis=-1),
    st.kurtosis(vibration3, axis=-1),
], axis=1)

if __name__ == '__main__':
    main()

```

After cleaning the data, the main Python script responsible for training and testing the RF Classification model was written. From sklearn, the RandomForestClassifier and MinMaxScaler packages were imported. The preprocessed data for unbalance levels 0 and 4 were loaded to test the model with data from the two extreme sides of the unbalance level spectrum:

```

train0 = np.load('0Dtsfresh.npy')[:, :, 0]
test0 = np.load('0Etsfresh.npy')[:, :, 0]
train1 = np.load('4Dtsfresh.npy')[:, :, 0]
test1 = np.load('4Etsfresh.npy')[:, :, 0]

```

The data was assembled into training and testing splits:

```

train_features = np.concatenate(
    [train0, train1], axis=0)
train_labels = np.concatenate(
    [np.zeros(train0.shape[0]),
     np.ones(train1.shape[0])])
test_features = np.concatenate(
    [test0, test1], axis=0)
test_labels = np.concatenate(
    [np.zeros(test0.shape[0]),
     np.ones(test1.shape[0])])

```

To normalize the data so that each column of values was comparable to the others, MinMaxScaler() was applied:

```

train_features = MinMaxScaler()\

```

```

    .fit_transform(train_features)
test_features = MinMaxScaler()\
    .fit_transform(test_features)

```

Finally, the RF model was created with specified hyperparameters and fitted to the training data:

```

model = RandomForestClassifier(
    max_depth=20,
    min_samples_leaf=1,
    n_estimators=300,
    random_state=0,
    verbose=True)
model.fit(train_features, train_labels)

```

Predictions were then generated for both the training and testing sets, and accuracy was computed for each.

VII. RESULTS AND ANALYSIS

The RF classification model was initially trained on the training subset of data and then tested on the testing subset of data. Accuracy, the metric used to evaluate the model, measures how often a machine learning model correctly predicts the intended outcome. It is calculated by dividing the number of correct predictions by the total number of predictions, and is expressed on a scale of 0 to 1 (or as a percentage). A model achieves a perfect accuracy of 1.0 when every prediction is correct.

After fitting and testing the RF classification model, the results are shown in Table I:

TABLE I
RF MODEL CLASSIFICATION ACCURACY

Dataset	Accuracy
Training set	100%
Testing set	100%

The RF classification model achieved 100% accuracy on both the training and testing subsets. While this may initially appear surprising, several factors provide a sound explanation for this result.

First, the model was trained and tested exclusively on unbalance levels “0” and “4”, the two most extreme ends of the unbalance spectrum. These two cases produce substantially different vibration signals, making the classification boundary particularly clear and easy for the model to learn. This is analogous to asking a classifier to distinguish between a completely stationary object and a violently vibrating one, which is a relatively straightforward task compared to differentiating intermediate levels. Had the model been tested on levels “1”, “2”, and “3”, where the differences in vibration signals are more subtle, accuracy would likely be lower.

Second, the large size of the dataset (roughly 10,000+ rows) allowed the RF model to learn the classification boundary with high precision. Random Forest models are well known to perform strongly on large structured datasets, particularly for binary classification tasks.

Third, the preprocessing script extracted highly discriminative statistical features, such as the standard deviation and kurtosis of each vibration sensor per batch. These features are well suited to capturing the difference between a balanced and unbalanced rotating shaft. This is further supported by the feature importance output, which showed that a small number of features carried the vast majority of the classification weight, indicating that the model identified clear and strong signals within the data.

The feature importances of the RF model were also examined by running:

```
print(model.feature_importances_)
```

The output is shown below, where each value represents the relative importance of a corresponding feature used by the RF model in its classification decisions:

```
Feature Importances:
[6.63498360e-03 2.05971150e-03 1.54364267e-02
 6.73537059e-04
 3.35922375e-03 1.77425579e-03 5.68760388e-05
 2.82125620e-05
 8.97288981e-04 6.02093922e-04 1.57146295e-04
 3.53294156e-05
 2.55571904e-03 1.74895357e-03 1.53635922e-04
 2.04588812e-05
 4.20016909e-04 ...
... 0.00000000e+00 0.00000000e+00 0.00000000e+00
0.00000000e+00 0.00000000e+00 0.00000000e+00
0.00000000e+00
1.70968510e-06 1.03511965e-06 0.00000000e+00
0.00000000e+00
1.66600185e-06 0.00000000e+00 4.28017266e-03
0.00000000e+00]
```

The most influential features have values on the order of 10^{-2} (e.g., $1.54364267e-02$), while the majority of features have negligible or zero importance. This confirms that only a small subset of the extracted features drove the classifier's decisions, and that these features captured a strong and consistent signal differentiating unbalance level "0" from level "4". The precise physical meaning of each feature index in relation to specific vibration characteristics remains an area for future investigation.

VIII. CONCLUSION AND FUTURE DIRECTIONS

In summary, the purpose of this research project was to determine the extent to which a machine learning classification model would be successful and effective in determining the magnitude of unbalance in a rotating shaft. Using vibration analysis data recorded experimentally by Mey et al. [12], a Random Forest classification model was implemented to classify vibration data by unbalance level. The model achieved 100% accuracy on both training and testing subsets, a result that is well supported by the nature of the experimental design: the model was evaluated on the two most extreme unbalance levels ("0" and "4"), which produce highly distinguishable vibration signals. Combined with the large dataset size and the effectiveness of the extracted statistical features, 100% accuracy on this binary classification task is a credible and expected outcome.

That said, this result should be interpreted with appropriate context. The high accuracy reflects performance on the easiest possible classification scenario: distinguishing no unbalance from maximum unbalance. It does not necessarily imply that the model would perform equally well across all five unbalance levels. Testing on intermediate levels "1", "2", and "3" remains an important next step to fully validate the model's effectiveness and generalizability.

If such a model were to be fully implemented in an industrial predictive maintenance setting, further modifications would also be required. Another tool would be needed to translate the RF's classifications into actionable decisions, such as flagging a rotating shaft for replacement or alerting factory operators when a part reaches a critical unbalance level.

Additionally, tracking the variance of the vibration output over time may prove valuable. A rising variance could serve as an early warning indicator that a shaft is degrading, complementing the classification output of the RF model. Understanding which specific tsfresh features contributed most significantly to the model's decisions could also yield further physical insight into the nature of rotating shaft unbalance, and is identified as a key direction for future work.

REFERENCES

- [1] A. Heng, S. Zhang, A. C. C. Tan, and J. Mathew, "Rotating machinery prognostics: State of the art, challenges and opportunities," *Mechanical Systems and Signal Processing*, vol. 23, no. 3, pp. 724–739, 2009.
- [2] Hudson Institute, Inc., "Mechanical Engineering in the 21st Century: Trends Impacting the Profession," American Society of Mechanical Engineers, 1999.
- [3] S. A. A. Hosseini and S. E. Khadem, "Free vibrations analysis of a rotating shaft with nonlinearities in curvature and inertia," *Mechanism and Machine Theory*, vol. 44, no. 1, pp. 272–288, 2009.
- [4] G. M. Sang, L. Xu, P. de Vrieze, and Y. Bai, "Predictive maintenance in Industry 4.0," in *Proc. 10th Int. Conf. Information Systems and Technologies*, ACM, 2020.
- [5] R. K. Mobley, *An Introduction to Predictive Maintenance*, 2nd ed. Butterworth-Heinemann, 2002.
- [6] C. Scheffer and P. Girdhar, *Practical Machinery Vibration Analysis and Predictive Maintenance*. Newnes, 2004.
- [7] M. Achouch et al., "On predictive maintenance in Industry 4.0: Overview, models, and challenges," *Applied Sciences*, vol. 12, no. 16, p. 8081, 2022.
- [8] P. Nunes et al., "Challenges in predictive maintenance – A review," *CIRP Journal of Manufacturing Science and Technology*, vol. 40, pp. 53–67, 2023.
- [9] Z. M. Cinar et al., "Machine learning in predictive maintenance towards sustainable smart manufacturing in Industry 4.0," *Sustainability*, vol. 12, no. 19, p. 8211, 2020.
- [10] A. Shafi, "Random Forest Classification with Scikit-Learn," DataCamp, Feb. 2023. [Online]. Available: <https://www.datacamp.com/tutorial/random-forests-classifier-python>
- [11] Jishnu, "Vibration Analysis on Rotating Shaft," Kaggle, Feb. 2022. [Online]. Available: <https://www.kaggle.com/datasets/jishnukoliyadan/vibration-analysis-on-rotating-shaft>
- [12] O. Mey, W. Neudeck, A. Schneider, and O. Enge-Rosenblatt, "Machine learning-based unbalance detection of a rotating shaft using vibration data," in *Proc. 25th IEEE Int. Conf. Emerging Technologies and Factory Automation (ETFA)*, Vienna, Austria, 2020, pp. 1610–1617.